

Black body radiation

Lying ≥ 4 eV above the ground state, Rydberg states are not populated thermally, except at very high temperatures. Accordingly, it is natural to assume that thermal effects are negligible in dealing with Rydberg atoms. However, Rydberg atoms are strongly affected by black body radiation, even at room temperature. The dramatic effect of thermal radiation is due to two facts. First, the energy spacings ΔW between Rydberg levels are small, so that $\Delta W < kT$ at 300 K. Second, the dipole matrix elements of transitions between Rydberg states are large, providing excellent coupling of the atoms to the thermal radiation. The result of the strong coupling between Rydberg atoms and the thermal radiation is that population initially put into one state, by laser excitation for example, rapidly diffuses to other energetically nearby states by black body radiation induced dipole transitions.^{1–3} Both the redistribution of population and the implicit increase in the radiative decay rates are readily observed. Although the above mentioned effects on level populations are the most obvious effects, the fact that a Rydberg atom is immersed in the thermal radiation field increases its energy by a small amount, 2 kHz at 300 K. While the radiation intensity is vastly different in the two cases, this effect is the same as the ponderomotive shift of the ionization limit in high intensity laser experiments.

Black body radiation

The most familiar way of representing black body radiation is to use the Planck radiation law for the energy density, or the square of the electric field, $\rho(\nu)$. Explicitly⁴

$$\rho(\nu) d\nu = \frac{8\pi h \nu^3}{c^3 (e^{h\nu/kT} - 1)}, \quad (5.1)$$

where k is the Boltzmann constant, h is Planck's constant, ν is the frequency of the black body radiation, and T is the temperature.

In Fig. 5.1 we show $\rho(\nu)$ vs ν at 300 K using both frequency and wavenumber as abscissae. A typical optical transition from the ground state of an atom has frequency $\nu = 3 \times 10^{14}$ Hz, and a transition between two Rydberg states has frequency $\nu \sim 3 \times 10^{11}$ Hz. Thus, it is apparent from Fig. 5.1 that to a ground state atom the black body radiation appears as a slowly varying, nearly static field, whereas to a Rydberg atom it appears to be a rapidly varying field.

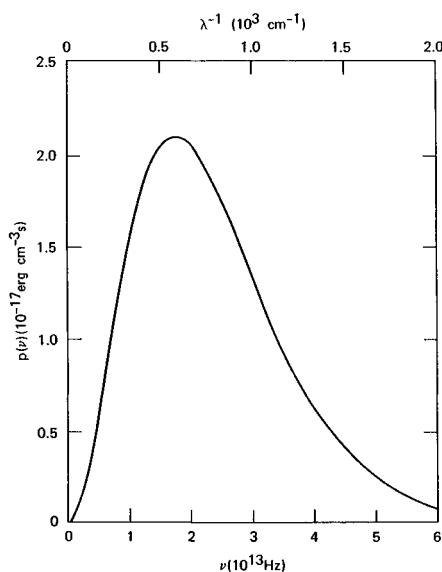


Fig. 5.1 Energy density $\rho(\nu)$ as a function of frequency ν at 300 K (from ref. 5).

While the representation of black body radiation given in Eq. (5.1) and Fig. 5.1 is the most familiar, it is not the most useful for the most important effect of black body radiation on Rydberg atoms, inducing transitions between neighboring levels.

To calculate the rates of these transitions it is more convenient to express the radiation field in terms of the number of photons per mode of the radiation field, i.e. the photon occupation number $\bar{n}$ of each mode. The photon occupation number $\bar{n}$ is given by⁴

$$\bar{n} = \frac{1}{e^{h\nu/kT} - 1}. \quad (5.2)$$

At low frequencies, for which $h\nu \ll kT$, Eq. (5.2) reduces to

$$\bar{n} \approx \frac{kT}{h\nu} \quad (5.3)$$

Fig. 5.2 shows the dependence of $\bar{n}$ on ν for $T = 300$ K, again with both frequency (Hz) and wavenumber (cm^{-1}) used as abscissae. It is useful to recall that at 300 K, $kT/h \approx 6 \times 10^{12}$ Hz or $kT/hc \approx 200$ cm^{-1} . This point is apparent in Fig. 5.2, since $\bar{n} = 1$ at $\lambda^{-1} \approx 200$ cm^{-1} . Since the vacuum fluctuations, which lead to spontaneous emission, are given by $\bar{n} = 1/2$, black body radiation at frequencies greater than $kT/h\bar{n}$, where $\bar{n} \ll 1$, does not lead to significant effects. For an atom in its ground state with transitions at $\sim 10^4$ cm^{-1} , black body induced transitions are unimportant, since $\bar{n} \ll 1$. However, for a Rydberg state with transitions at 10 cm^{-1} , where $\bar{n} \sim 10$, the black body induced transition rates can

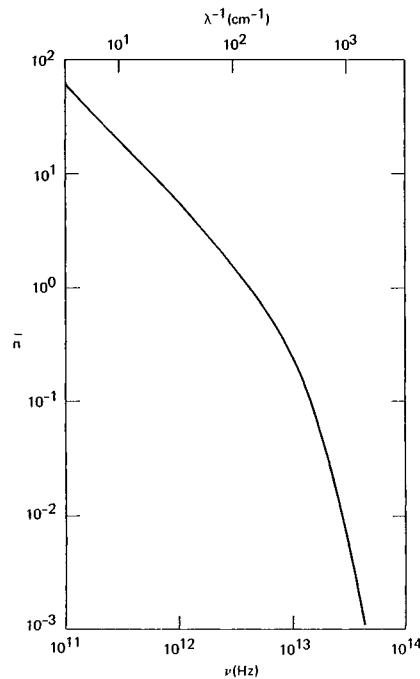


Fig. 5.2 Occupation number $\bar{n}$ as a function of frequency ν at 300 K (from ref. 5).

be an order of magnitude larger than the spontaneous emission rates, and the effect of black body radiation is significant.

Eqs. (5.1)–(5.3) present black body radiation in a familiar form. Both to conform to the general use of atomic units in this book and to simplify the calculation of the Rydberg atoms' response to the radiation we reexpress Eqs. (5.1)–(5.3) in atomic units.⁵

We reexpress Eq. (5.1) as

$$\rho(\omega)d\omega = \frac{2\alpha^3\omega^3d\omega}{\pi(e^{\omega/kT} - 1)}, \quad (5.4)$$

where α is the fine structure constant and ω is the energy in au. Similarly, we may reexpress Eq. (5.2) as

$$\bar{n} = \frac{1}{e^{\omega/kT} - 1} \quad (5.5)$$

and Eq. (5.3) for $\omega \ll kT$ as

$$\bar{n} \approx \frac{kT}{\omega}. \quad (5.6)$$

We now consider the transition rates for absorption of and stimulated emission induced by the black body radiation and compare these rates to the spontaneous

emission rates. This comparison provides a reasonable indication of the importance of the effects of black body radiation induced transitions for Rydberg atoms.⁶

Black body induced transitions

As shown in the previous chapter, the spontaneous decay rate of the $n\ell$ state to the lower lying $n'\ell'$ state is given by the Einstein A coefficient $A_{n'\ell',n\ell}$.⁷ In the presence of thermal radiation the stimulated emission rate $K_{n'\ell',n\ell}$ is simply $\bar{n}$ times as large as the spontaneous rate. Explicitly,

$$K_{n'\ell',n\ell} = \bar{n}A_{n'\ell',n\ell}, \quad (5.7)$$

where $\bar{n}$ is implicitly evaluated at the frequency $\omega_{n'\ell',n\ell}$. It is convenient to reexpress Eq. (5.7) in terms of the average oscillator strength,

$$K_{n'\ell',n\ell} = -2\bar{n}\alpha^3\omega_{n'\ell',n\ell}^2\bar{f}_{n'\ell',n\ell}, \quad (5.8)$$

where α is the fine structure constant, $\omega_{n'\ell',n\ell}$ is the energy difference $W_{n'\ell'} - W_{n\ell}$, and

$$\bar{f}_{n'\ell',n\ell} = \frac{2}{3}\omega_{n'\ell',n\ell}\frac{\ell_{\max}}{2\ell+1}|\langle n'\ell'|r|n\ell\rangle|^2, \quad (5.9)$$

where $\ell_{\max}$ is the larger of ℓ and ℓ' .

The black body photons can also be absorbed as the atoms in the $n\ell$ state make the transition to a higher lying $n'\ell'$ state. Both the stimulated emission and absorption rates are given by

$$K_{n'\ell',n\ell} = 2\bar{n}\alpha^3\omega_{n'\ell',n\ell}^2|\bar{f}_{n'\ell',n\ell}|. \quad (5.10)$$

Because of the variation of $\bar{n}$ with frequency, the frequency dependence of $K_{n'\ell',n\ell}$ is quite different from that of $A_{n'\ell',n\ell}$, and these two processes favor different final states. This point is made graphically in Fig. 5.3, which is a plot of $A_{n'p,18s}$ and $K_{n'p,18s}$ vs n' for the Na 18s state and $T = 300$ K. As shown in Fig. 5.3, black body radiation favors transitions to nearby states and spontaneous emission favors transitions to the lowest lying states.

In addition to driving transitions to discrete states the black body radiation can also photoionize the Rydberg atoms. The photoionization rate, $1/\tau_{n\ell}^P$ is given by⁸

$$\frac{1}{\tau_{n\ell}^P} = \frac{2\pi}{3} \int_{1/2n^2}^{\infty} \left\{ \frac{\ell}{2\ell+1} |\langle n\ell|r|\varepsilon\ell-1\rangle|^2 + \frac{(\ell+1)}{2\ell+1} |\langle n\ell|r|\varepsilon\ell+1\rangle|^2 \right\} \rho(\omega) d\omega. \quad (5.11)$$

The matrix elements are the radial matrix elements of r between the initial $n\ell$ state and continuum $\ell-1$ and $\ell+1$ waves which are normalized per unit energy.

Just as we previously summed the spontaneous emission rates over the possible final states to obtain the total spontaneous decay rate $1/\tau_{n\ell}$, we can sum the black

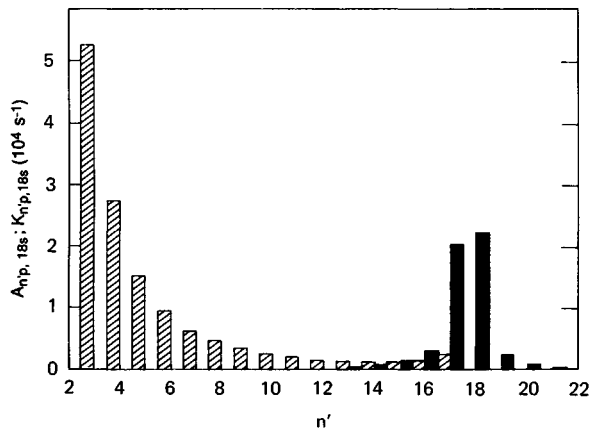


Fig. 5.3 Spontaneous emission rates of the 18s state to lower lying $n'p$ states, $A_{n'p, 18s}$, ($\square$) and the 300 K black body transition rates of the 18s state to higher and lower lying $n'p$ states, $K_{n'p, 18s}$, ($\blacksquare$) as a function of n' (from ref. 5).

body radiation transition rates to compute the total black body radiation induced decay rate $1/\tau_{n\ell}^{\text{bb}}$. Explicitly,

$$\frac{1}{\tau_{n\ell}^{\text{bb}}} = 2\alpha^3 \sum_{n'} \bar{n} \omega_{n'e', n\ell}^2 |f_{n'e', n\ell}| + \frac{1}{\tau_{n\ell}^{\text{P}}}, \quad (5.12)$$

and the total decay rate $1/\tau_{n\ell}^{\text{T}}$ is given by

$$\frac{1}{\tau_{n\ell}^{\text{T}}} = \frac{1}{\tau_{n\ell}} + \frac{1}{\tau_{n\ell}^{\text{bb}}}. \quad (5.13)$$

From Fig. 5.3 it is apparent that the most important contributions to the black body radiation decay rate are from transitions to nearby states for which $|\omega_{n'e', n\ell}| < kT$. In this case we may substitute Eq. (5.6) for $\bar{n}$ and write $1/\tau_{n\ell}^{\text{bb}}$ as

$$\frac{1}{\tau_{n\ell}^{\text{bb}}} = 2\alpha^3 kT \sum_{n'} \omega_{n'e', n\ell} \bar{f}_{n'e', n\ell}. \quad (5.14)$$

In Eq. (5.14) the summation over n' implicitly includes the continuum. For a one-electron atom we may use the sum rule⁷

$$\sum \omega_{n'e', n\ell} \bar{f}_{n'e', n\ell} = \frac{2}{3n^2}, \quad (5.15)$$

which we may use to write

$$\frac{1}{\tau_{n\ell}^{\text{bb}}} = \frac{4\alpha^3 kT}{3n^2}. \quad (5.16)$$

For $T = 300 \text{ K}$, Eq. (5.16) is accurate to 30% for $n > 15$, and as n is increased Eq. (5.16) becomes an increasingly good approximation. More important, it brings

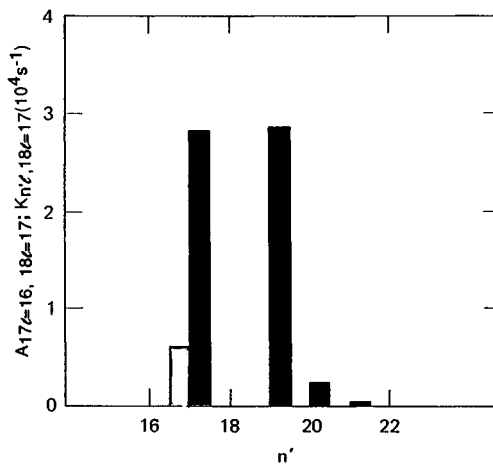


Fig. 5.4 Spontaneous emission rate of the $n = 18, \ell = 17$ state to the $n = 17, \ell = 16$ state, $A_{17\ell=16, 18\ell=17}$, ($\square$) and the 300 K black body transition rates of the $n = 18, \ell = 17$ state to $n'\ell'$ states of $\ell' = \ell \pm 1$, $K_{n'\ell', 18\ell=17}$, ($\blacksquare$) as a function of n' (from ref. 5).

out two interesting points. First, the total black body decay rate $1/\tau_{n\ell}^{\text{bb}}$ does not depend on ℓ , only on n . Since the radiative decay rate decreases rapidly as ℓ is increased,⁹ for high ℓ states the black body induced decay rate is often much larger than the spontaneous emission rate. This point is shown graphically in Fig. 5.4 in which the spontaneous emission rates and black body radiation emission and absorption rate are shown for the $n = 18, \ell = 17$ state. In contrast to the 18s state, for which $1/\tau^{\text{bb}} \sim 0.2/\tau$, the $n = 18, \ell = 17$ state has a black body decay rate which is ten times larger than its spontaneous decay rate. Second, because of the $1/n^2$ scaling of the black body radiation decay rate, it is evident that even for low ℓ states that at high enough n the black body radiation rate exceeds the spontaneous emission rate, which scales as $1/n^3$. Even at $n = 20$, the black body radiation rate is at least 20% of the spontaneous emission rate for all ℓ states, and in many experiments it is a significant source of population redistribution.

Black body energy shifts

That part of the black body spectrum coincident with the atomic transition frequencies leads to the transitions which redistribute the population. In contrast, all the energy of the black body radiation contributes to the shift of the energy levels. The energy shift is a second order ac Stark shift, and for state n the shift ΔW_b is given by¹⁰

$$\Delta W_{n\ell} = \sum_{n'\ell'} \int_0^\infty \frac{|\langle n'\ell' | r | n\ell \rangle|^2 E_{\omega_b}^2 \omega_{n\ell, n'\ell'}}{2(\omega_{n\ell, n'\ell'}^2 - \omega_b^2)} d\omega_b \quad (5.17)$$

where E_{ω_b} is the electric field of the black body radiation in a bandwidth $d\omega_b$ at frequency ω_b . Since $E_{\omega_b}^2$ is proportional to the energy density, examining Fig. 5.2 we can see that at room temperature the peak of the black body radiation energy density is at $\sim 500 \text{ cm}^{-1}$, far above the strong $n \sim 20$ transitions which have wavenumbers of $\sim 20 \text{ cm}^{-1}$. In fact, for atoms in Rydberg states it is often true that the frequencies of the strong transitions are much lower than the frequency of the black body radiation, i.e. $\omega_{n\ell, n'\ell'} \ll \omega_b$, in which case we can ignore $\omega_{n\ell, n'\ell'}$ in the denominator of Eq. (5.17) and rewrite $\Delta W_{n\ell}$ as

$$\Delta W_{n\ell} = \left(\sum_{n'\ell'} \bar{f}_{n'\ell', n\ell} \right) \left(\int_0^\infty \frac{E_{\omega_b}^2 d\omega_b}{4\omega_b^2} \right). \quad (5.18)$$

Using the oscillator strength sum rule⁷

$$\sum_{n'\ell'} \bar{f}_{n'\ell', n\ell} = 1 \quad (5.19)$$

and integrating over ω_b in Eq. (5.18), we may express Eq. (5.18) as

$$\Delta W_{n\ell} = \frac{\pi}{3} \alpha^3 (kT)^2. \quad (5.20)$$

Eq. (5.20) is accurate to $\sim 10\%$ for $n > 15$ at $T = 300 \text{ K}$. Eq. (5.20) indicates that all Rydberg states experience the same energy or frequency shift, which is $+2.2 \text{ kHz}$ at 300 K . Eq. (5.20) corresponds to the energy shift of a free electron in an oscillating electric field. This point is more apparent if we simply use the oscillator strength sum of Eq. (5.19) and apply it to the case of a monochromatic field of angular frequency ω_b for which $\int_0^\infty E_b^2 d\omega_b = E^2$. In this case Eq. (5.18) can be written

$$\Delta W_{n\ell} = \frac{E^2}{4\omega^2}, \quad (5.21)$$

which is the average kinetic energy of a free electron in a field $E \cos \omega t$. The equality of the energy shift of a free electron and an electron in a Rydberg state is not surprising since the physical effect of the ac black body field, at a frequency high compared to the orbital frequency of the electron, is to superimpose a fast, $\sim \omega_b$, wiggle on the orbital motion of the electron. The energy of the wiggle motion is independent of the much slower orbital velocity.

While the black body radiation energy is mostly at frequencies high compared to the Rydberg state frequencies, it is low compared to the frequencies of transitions of low lying states. Explicitly, $\omega_{n'\ell', n\ell} \gg \omega_b$, and for low lying states we can ignore ω_b in the denominator of Eq. (5.18). In this case the Stark shift is equal to that produced by a static field and can be expressed as

$$\Delta W_{n\ell} = \sum_{n'\ell'} - \frac{\bar{f}_{n'\ell', n\ell}}{\omega_{n'\ell', n\ell}^2} \int_0^\infty \frac{E_{\omega_b}^2 d\omega_b}{4}. \quad (5.22)$$

For a one-electron Rydberg atom we may use the hydrogenic sum rule¹¹

$$\sum_{n'} \frac{\bar{f}_{n'\ell',n\ell}}{\omega_{n'\ell',n\ell}^2} = \frac{9}{2} \quad (5.23)$$

so that we may write Eq. (5.22) as

$$\Delta W_n = - \int \frac{9}{8} E_{\omega_b}^2 d\omega_b = - \frac{3}{5} (\alpha\pi)^3 (kT)^4. \quad (5.24)$$

Evaluation of Eq. (5.24) gives a shift of -0.036 Hz at 300 K, which is negligible for all practical purposes.

Finally, we must remember that for some states it will be true that $\omega_{n'\ell',n\ell} \sim \omega_b$, in which case neither of the above approximations is valid and Eq. (5.17) must be evaluated explicitly as has been done by Farley and Wing.¹² At 300 K $\omega_{n\ell,(n+1)\ell'} \sim \omega_b$ for $n \sim 8$.

If we confine our attention to Rydberg states of $n > 15$, for which Eq. (5.20) is valid, it is apparent that all the Rydberg states and the ionization limit are shifted up in energy by an amount which is proportional to the square of the temperature. If the atoms are exposed to a monochromatic radiation field satisfying the same frequency criteria, the Rydberg state energies and the ionization limit are shifted according to Eq. (5.21), which is often termed the ponderomotive energy shift.¹³

In this chapter we have implicitly assumed the Rydberg atom to be a one electron-atom. In the perturbed Rydberg series of, for example, alkaline earth atoms, Rydberg states can have mixed valence–Rydberg character. In such states the black body effects are reduced by a factor equal to the fractional Rydberg character.¹⁴

Initial verification

The experimental observations which called attention to the effects of room temperature black body radiation on Rydberg atoms were made using Rydberg states of Xe and Na.^{1–3} However, Pimbert had earlier noted the effect of higher temperatures.¹⁵ Two kinds of experiments were performed, measurements of population redistribution and increased decay rates. Both of the effects measured could conceivably arise from collisions, and this possibility was systematically ruled out in the experiments. In the Xe experiment atoms in a metastable beam were excited to the 26f state by a pulsed laser and exposed to the thermal radiation for periods of 1.5, 7.5, and 15.5 μ s, after which a ramped field was applied to the atoms. This procedure leads to the field ionization signals shown in Fig. 5.5.¹ For each time delay there is a clear progression of peaks in the field ionization spectra which are assigned the n values shown. At longer delay times the intensities of the

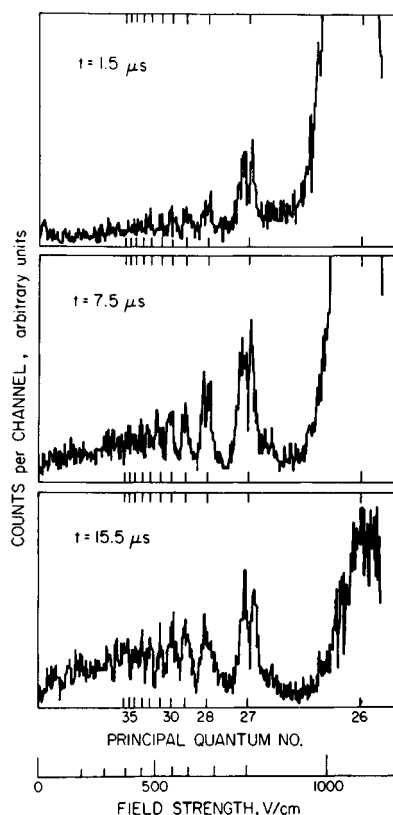


Fig. 5.5 Field ionization spectra at different times after the laser pulse excites Xe atoms to the 26f level showing the black body radiation induced transfer to higher levels (from ref. 1).

$n > 26$ peaks grow, as would be expected for progressively longer exposure times to the thermal radiation. In addition, due to the fact that the black body radiation induced transitions are dipole transitions, only d and g final states are expected. The fact that the field ionization spectra exhibit a sharp peak for each n is consistent with only one or two, not all, ℓ states being populated by the thermal radiation.¹ If all the ℓ states were populated there would be no sharp peaks in Fig. 5.5.

In an analogous experiment in which the initial population was placed in the Na 18s state, the transferred population was found to be in the higher lying p states, as expected for dipole transitions. Furthermore, the fraction of the atoms which had undergone transitions to each of the higher p states after 5 μ s was in good agreement with values calculated on the basis of 300 K black body radiation induced transitions, as shown by Table 5.1.²

The other initial experiment was the observation of the increase in the decay rate due to 300 K black body radiation, shown by Eq. (5.13).³ The decay rate of

Table 5.1. *Calculated and observed populations in higher lying np states 5 μ s after the initial population of the Na 18s state expressed in terms of the initial 18s population.*^a

Final state	Calculated yield	Observed yield
18p	4.2%	5.0%
19p, 20p	1.6%	1.6%
>20p, ϵ p	1.0%	1.2%
Total	6.8%	7.8%

^a(see ref. 2)

Table 5.2. *Lifetimes of the Na 17p and 18p states.*^a

State	τ (μ s)	τ^{bb} (μ s)	τ^{T} (calc) (μ s)	τ^{T} (obs) (μ s)
17p	48.4 ^b	22.7	15.5	11.4 ^{+5.6} _{-1.4}
18p	58.4 ^b	25.6	17.9	13.9 ^{+8.8} _{-2.9}

^a(from ref. 3)

^b(from ref. 17)

the Na 18s state at 300 K differs from its 0 K rate by about 20%, which is not a large enough difference that a measurement at 300 K alone is entirely convincing. The Na np states of $n \sim 20$, on the other hand, have long lifetimes^{16,17} and offer a reasonable prospect for a convincing measurement, even if made only at 300 K. The Na 17p and 18p lifetimes were measured by exciting Na atoms in a beam to the 17p and 18p states and observing the populations in these states as a function of time after the laser pulse. The results of the experiment are given in Table 5.2, which shows clearly that the experimentally observed values are in good agreement with the lifetimes calculated with the black body radiation, but in stark disagreement with the 0 K lifetimes.

Temperature dependent measurements

Subsequent to the initial 300 K observations, measurements were done as a function of temperature. The first measurements, by Koch *et al.*,¹⁸ were done at

elevated temperatures, and later measurements were done at temperatures below room temperature. A critical issue in such measurements is knowing the temperature the atoms are experiencing. Typically an enclosure cooled to a temperature from 6 K to 300 K surrounds the atoms, shielding them from the 300 K radiation from the walls of the vacuum chamber. However, simply making an enclosure in which 90% of the solid angle seen by the atoms is cold is not adequate, because at far infrared wavelengths many materials, and virtually all metal surfaces, are excellent reflectors. Thus a 10% aperture to the warm world outside is adequate to raise substantially the temperature the atoms in the enclosure experience.

Two approaches have been used to ensure that the temperature experienced by the atoms is close to the temperature of the enclosure. Hildebrandt *et al.* lined the inside of the enclosure with graphite coated Cu wool, which they verified separately to be an effective way of absorbing far infrared radiation.¹⁹ An alternative approach was followed by Spencer *et al.*²⁰ They covered the apertures through which laser beams propagated with glass, which is opaque at far infrared wavelengths, and those through which the atomic beam propagated with a fine, 70 μm , mesh, which blocks all radiation with $\lambda > 140 \mu\text{m}$.

Experiments with Xe analogous to the previously described Xe experiment were carried out at both 90 K and 300 K. Specifically the time dependences of the populations in higher nd and ng states were measured subsequent to population of an nf state by a pulsed laser.¹⁹ The time dependences of the observed populations were fit to a model which yielded the radiative transfer rates from the initially excited state to the final states. Not surprisingly, as the temperature was reduced from 300 K to 90 K, not only was the overall radiative transfer reduced, but the transfer to the highest lying states was most sharply reduced, as expected from the dependence of the photon occupation number on frequency and temperature. For example the transfer from the Xe 25f to 26d,g states was reduced by a factor of 3 while the 25f–27d,g transfer was reduced by a factor closer to 5.

Spencer *et al.* populated the Na 19s state and observed the sum of the populations in the 19s, 19p, and 18p states, at times from 2–32 μs after the laser excitation, to determine the sum of the radiative transition rates from the 19s to 18p and 19p states.²⁰ The measurements were made at temperatures from 6 K to 210 K with the results shown in Fig. 5.6. As shown by Fig. 5.6, the observed rates agree well with the calculated rates, shown by the line. At 0 K the rate is entirely due to the spontaneous 19s $\rightarrow$ 18p transition, and the increase above this rate is due to thermal radiative transfer to both the 18p and 19p states. It is interesting to note that above ~ 30 K the transfer rate increases linearly with T as expected from Eq. (5.16). The linear behavior only occurs above 30 K since only then is $kT > \omega$ for these two transitions.

While the Na 19s $\rightarrow$ 18p, 19p transfer rate is an excellent illustration of the validity of Eq. (5.16) for $\omega < kT$, the photoionization of Rydberg atoms of $n \approx 20$ by black body radiation is a test of the regime $\omega \sim kT$, and this regime has been explored by Spencer *et al.*²¹ Specifically, they measured the relative rates for

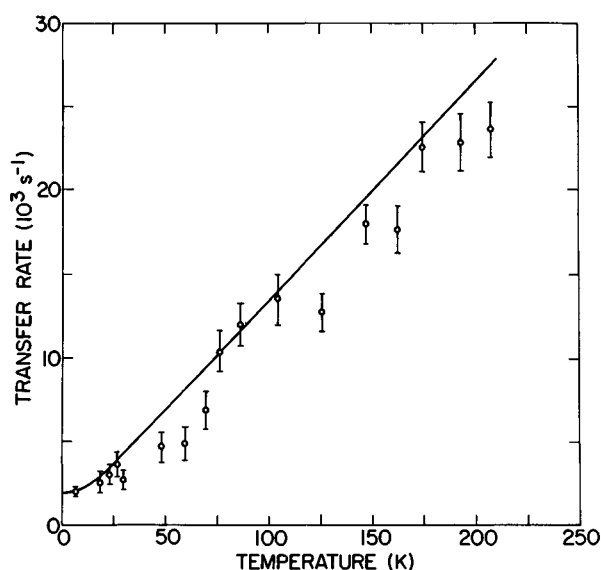


Fig. 5.6 Experimental and theoretical transfer rate of the Na 19s state to the 18p and 19p states vs temperature. The solid line is calculated with no adjustable parameters (from ref. 20).

photoionization of the Na 17d state by thermal radiation at temperatures of 90–300 K. The experiments were done by exciting the Na 17d state, allowing photoionization to occur for 500 ns, a time ten times shorter than the 17d lifetime, and then collecting the photoions by applying a small field pulse of 8 V/cm. A field this small was verified to produce negligible field ionization of bound states excited by black body radiation. When the relative rates are normalized to the calculated rate at 300 K, they are found to be in excellent agreement with the calculated rates.

Since the 17d state is bound by 380 cm^{-1} , which exceeds kT , the photoionization rate varies rapidly with temperature, roughly a factor of 100 between 90 K and 300 K, as shown by Fig. 5.7. The rapid, nearly exponential, dependence of the photoionization rate on temperature shown in Fig. 5.7 is to be contrasted with the linear temperature dependence of the $19s \rightarrow 19p$, $18p$ transfer rate shown in Fig. 5.6.

Suppression and enhancement of transition rates

We have described the effects of black body radiation in free space. In a closed cavity the radiation is confined to the allowed modes of the cavity. In essence all the thermal radiation is forced into the cavity modes, raising the intensity at the

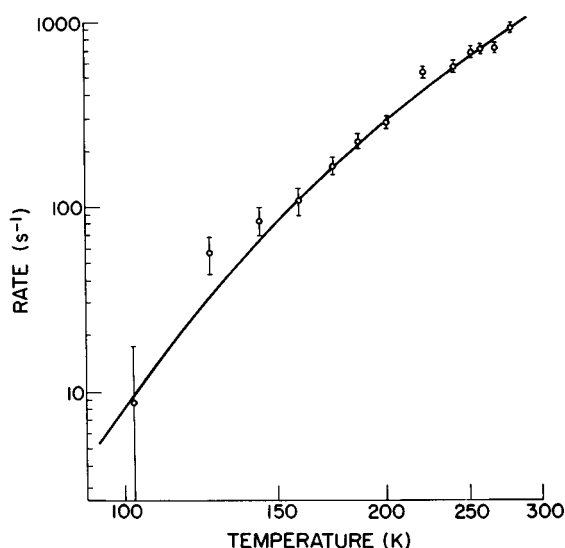


Fig. 5.7 Black body induced photoionization rate vs temperature for the 17d state in Na. Scales are logarithmic. The solid line represents the calculated values. Experimental points are normalized to the calculated value at 300 K (from ref. 21).

frequencies of the modes by a factor of Q relative to the free space value.²² Here Q is the quality factor of the cavity, the frequency of a mode divided by its linewidth (FWHM), i.e. $Q = \nu/\Delta\nu$. Similarly, between the cavity modes the intensity is substantially depressed. At a cavity mode resonance the radiation transfer rate is increased, by a factor of Q , and away from a resonance, the transfer rate is suppressed. As pointed out by Kleppner, Rydberg atoms, with their long wavelength transitions, provide a nearly ideal system for the study of such notions.²³

The first experimental realization of this notion was by Vaidyanathan *et al.* who excited Na atoms to nd states between two parallel plates 0.337 cm apart.²⁴ The region between these plates is cutoff for propagation of radiation of wavelengths longer than 0.674 cm polarized parallel to the plate surfaces, and such radiation is not present between the plates. Stated another way, frequencies less than the cutoff frequency $\nu_c = 1.48 \text{ cm}^{-1}$ do not propagate. Radiation polarized perpendicular to the surfaces can propagate freely between the plates. The zero static field 29d–30p interval is just under 1.48 cm^{-1} . However, the application of a small static field increases the separation of the levels, and a spacing of 1.48 cm^{-1} occurs at a static field of 2.4 V/cm. Thus as the static field is increased from zero to beyond 2.4 V/cm the 29d $\rightarrow$ 30p transition frequency passes from below to above cutoff, and the black body radiation transition rate increases accordingly.

In the experiment Na atoms are excited by two lasers from the ground state to the nd state in an environment cooled to 180 K at which point $\bar{n} = 86$ for the 29d $\rightarrow$ 30p transition. The atoms are exposed to the thermal radiation for 20 μs ,

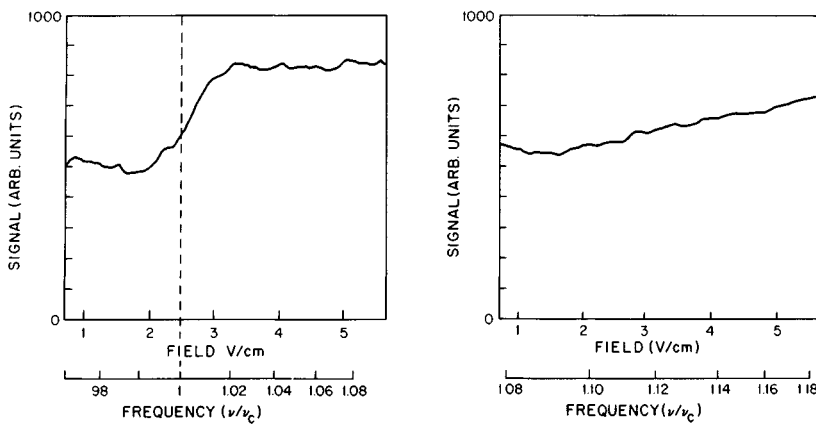


Fig. 5.8 Black body radiative transfer signals in Na located between parallel conducting plates for $29d \rightarrow 30p$ (left-hand side) and $28d \rightarrow 29p$ (right-hand side) as a function of the absorption frequency. The cutoff frequency is $\nu_c = 1/2d = 1.48 \text{ cm}^{-1}$, where d is the plate separation. The increase in the transfer rate at $\nu = \nu_c$ (left-hand side) is due to the “switching on” of the radiation polarized parallel to the plates (from ref. 24).

after which they are exposed to a field ionization pulse to ionize atoms in the $(n + 1)p$ state selectively. In Fig. 5.8 we show the results obtained when the $28d$ and $29d$ states are excited as functions of static field over the range $0\text{--}5 \text{ V/cm}$. As shown in Fig. 5.8, the $29d\text{--}30p$ interval is tuned through the cutoff frequency of the plates in this field range. Correspondingly, there is a sharp increase in the population found in the $30p$ state when the field tunes the $29d\text{--}30p$ interval through the cutoff frequency. In contrast, the $28d\text{--}29p$ interval always exceeds the cutoff frequency, and no sharp increase in the $29p$ population is observed as the field is tuned.

Just as the black body transitions can be suppressed, they can also be enhanced, by tuning to a cavity resonance. Raimond *et al.*²⁵ have passed a thermal beam of Na atoms excited to the $30s_{1/2}$ state through a 300 K Fabry–Perot cavity, which can be tuned to the $30s_{1/2}\text{--}30p_{1/2}$ resonance at 134 GHz , and detected the atoms downstream by field ionization. With the cavity tuned off resonance 5% of the atoms make the $30s_{1/2} \rightarrow 30p_{1/2}$ transition, but when it is tuned to resonance an additional 5% make the transition, in agreement with the calculated value.²⁵

In any low angular momentum state the radiative decay rate is usually dominated by the high frequency transitions to low lying states, and as a result it is impossible to control completely the decay rate using a millimeter wave cavity. In a circular $\ell = m = n - 1$ state the only decay is the far infrared transition to the $n - 1$ level, and Hulet *et al.* have observed the suppression of the decay of this level.²⁶ They produced a beam of Cs atoms in the circular $n = 22$, $\ell = m = 21$ state by pulsed laser excitation and an adiabatic rapid passage technique.²⁷ The beam of circular state atoms then passed between a pair of plates 6.4 cm wide, 12.7 cm long, spaced by $230.1 \mu\text{m}$, and held at 6 K . The 0 K radiative lifetime is $460 \mu\text{s}$, and

the transit time of the atoms through the plates to the detector is approximately one lifetime at the peak of the velocity distribution.

In zero static field the wavelength λ of the $n = 22, \ell = 21 \rightarrow n = 21, \ell = 20$ transition is $450 \mu\text{m}$, and $\lambda = 0.98(2d)$ where d is the plate spacing. However, using the plates to apply a static field it is possible to reduce continuously the energy separation between the levels by up to 4%, increasing the wavelength of the transition to $1.02(2d)$. In other words it is possible to increase the wavelength of the transition from below cutoff to above cutoff by varying the tuning field. At this point it is worth recalling that in the static field the $\ell = 21, m = 21$ circular $n = 22$ state must interact with radiation polarized parallel to the field plates to decay to the $n = 21, \ell = m = 20$ state. The radiation polarized perpendicular to the plates cannot drive this transition. When the wavelength is longer than the cutoff wavelength, i.e. $\lambda > 2d$, radiative decay is suppressed, i.e. $\tau_{\text{cav}} = \infty$, but when the wavelength is less than the cutoff wavelength $\lambda < 2d$ the radiative decay rate is increased by 50%.²⁶

They recorded time of flight spectra of atoms remaining in the circular state as a function of the spacing of the plates and the tuning field applied, and from these spectra determined the decay rates. Decay rates were measured and calculated for three conditions. First the plate spacing was increased by a factor of 30 to observe the free space decay. They observed a lifetime of $450(10) \mu\text{s}$, in good agreement with the calculated 6 K value of $451 \mu\text{s}$. With the plate spacing returned to $230.1 \mu\text{m}$ the decay rates were measured with the electric field set so as to tune the wavelength both below and above the cutoff wavelength. With a low electric field, for which $\lambda < 2d$, the measured decay rate was 50% larger than the free space value, to within 5%. When the field was increased so that $\lambda > 2d$, roughly twice the number of atoms was detected and the decay rate was consistent with zero, to within 5% of the free space decay rate.

Level shifts

Rydberg states which are pure Rydberg states, not containing admixtures of valence states, all have the same positive energy shift in a thermal radiation field. To measure this shift requires careful measurement of a transition from a long lived low lying atomic state. Holberg and Hall have measured the thermal shift of Rb atoms exposed to a thermal source which ranged in temperature from 350 K to 1000 K using two photon, Doppler free spectroscopy between the ground state and the Rydberg states.²⁸ In their experiment a Rb beam passes through the two arms of a folded optical cavity, so the atoms interact with the optical radiation twice, providing a Ramsey interference pattern. An attraction of the Ramsey method is that the Ramsey interference pattern is located at the optical field free location of the $5s \rightarrow ns$ transition. It is thus unaffected by laser

light shifts. The central Ramsey interference fringe is 40 kHz wide, the inverse of the transit time of the atoms between the two arms of the optical cavity. The radiation from the heated black body source is focused on the atomic beam between its intersections with the two laser beams. Due to the small solid angle subtended by the heated black body source at the intersection of the atomic and laser beams, it is only $\sim 10\%$ as effective as a complete enclosure at the same temperature. The radiation from the heated black body source is chopped and the position of the central fringe of the Ramsey pattern observed with and without the thermal radiation. In spite of the reduced efficiency of the source in producing a shift, blue shifts of up to 1.4 kHz were observed, at 876 K. The blue shift increases with temperature as predicted by Eq. (5.20), and while the results do not provide a stringent test, they are clearly consistent with Eq. (5.20).

The black body shifts are not confined to Rydberg atoms, but also alter the frequency of atomic clock transitions. Itano *et al.* have shown that the Cs 9 GHz ground state hyperfine interval, the definition of the second, is increased by one part in 10^{14} when the temperature is raised from 0 K to 300 K.²⁹

Experimental manifestations and uses

As we have already pointed out, the presence of black body radiation invariably decreases the lifetime of a Rydberg state, and the decreased lifetimes have been reported in many systems.^{3,30–32} What is most important, though, is the transfer of population to nearby states, and this effect must be taken into account in making measurements. To illustrate this point we consider the measurement of the 300 K lifetime of a Rydberg state using selective field ionization, an apparently straightforward measurement.⁶ The 18s state of Na is calculated to have a 0 K radiative lifetime of 6.37 μs , which is reduced to 4.87 μs by the 300 K black body radiation. Na atoms were excited to the 18s state at time $t = 0$, and the population remaining at later times was determined by applying an electric field pulse high enough to ionize easily the atoms in the 18s state. This measurement yielded an apparent lifetime of 7.8 μs , a value longer than the 0 K radiative lifetime. The problem was that the field also ionized all the long lived, np states of $n \geq 17$ to which population was transferred by black body radiation. Only when the signal from higher lying states was detected separately was it possible to correct the observed 18s signal for the presence of the long lived np states. The final value was 4.78 μs , in excellent, and probably fortuitous, agreement with the calculated value of 4.87 μs .

As shown by Table 5.2, at 300 K the $n = 17$ and $n = 18$ black body decay rates are $\approx 5 \times 10^4 \text{ s}^{-1}$, which might seem tolerable. However, in even slightly dense

samples thermal radiation can trigger superradiance, which depopulates an initially populated state in 10 ns.^{33,34} For superradiance to occur between two levels (crudely speaking, laser action without mirrors), a necessary condition is that the gain, G , be 1 along the length of the sample. This criterion may be expressed as³³

$$G = NL\sigma \geq 1, \quad (5.25)$$

where N is the difference in population densities of the upper and lower levels, L is the length of the sample, and σ , the optical cross section, is given by³⁵

$$\sigma = \frac{g_\ell \lambda^2 A_{u\ell}}{8\pi g_u \Delta}, \quad (5.26)$$

where g_ℓ and g_u are the degeneracies of the upper and lower states. $A_{u\ell}$ is the Einstein coefficient for the transition, and Δ is the linewidth of the transition, to which the main contributions are usually radiative and Doppler broadening. Because of the long wavelength of the transitions between Rydberg states, the above gain criterion for superradiance can be met with a very low density of atoms, hundreds of atoms in a volume of 10^{-3} cm^{-3} , and is easily met even in a low density atomic beam. The mere presence of adequate gain does not, however, ensure superradiance. Some initial photons are required to trigger the superradiance, and due to the low spontaneous transition rates of the long wavelength transitions between Rydberg states, they are unlikely to come from spontaneous emission. On the other hand, the omnipresent black body radiation easily triggers superradiance.

Gounand *et al.* have used superradiant cascades initiated by black body radiation to populate efficiently states which would normally be inaccessible using optical excitation.³⁴ In vapor cell experiments they observed the population in Rb Rydberg states to undergo several very rapid superradiant cascades to populate lower lying levels. The population is transferred in times far shorter than would be possible by simple radiative decay.³⁴ Fig. 5.9 shows the population in various Rb Rydberg states after the initial population of the 12s state. Normal radiative decay would require $\sim 3 \mu\text{s}$ to populate the 7f state, but as shown in Fig. 5.9, only 20 ns is required when superradiant cascades occur.

Stoneman and Gallagher have used black body radiation to make precise measurements of the avoided crossings between the Kn s and $n - 2$ Stark manifold states in electric fields,³⁶ and these measurements are described in Chapter 6.

Far infrared detection

An interesting potential application of Rydberg atoms is as a far infrared (or microwave) detector, a notion first suggested by Kleppner and Ducas.³⁷ The basic

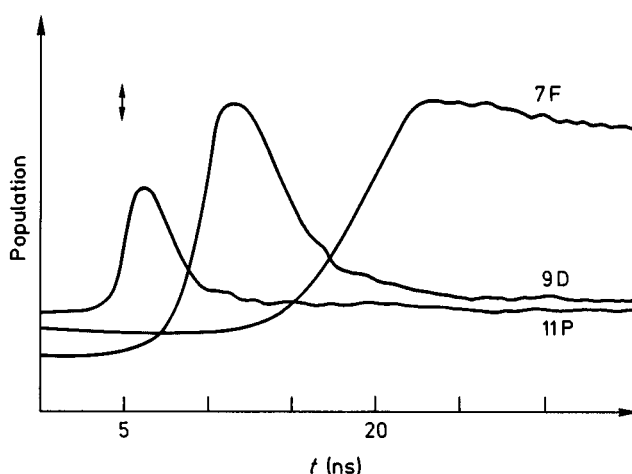


Fig. 5.9 Time dependence of the population in Rb Rydberg states after the initial population of the 12s state showing the rapid superradiant cascades (from ref. 35).

idea of the most straightforward type of device is to use a laser to make a target of Rydberg state atoms in one state, *A*. This target is exposed to the radiation to be detected at the infrared frequency ω_{AB} , which is equal to the frequency of the atomic transition from *A* to a higher energy state *B*. If the density of atoms in state *A* in the target is high enough that the target is optically thick to the radiation at frequency ω_{AB} , a photon at this frequency will be absorbed with nearly unit probability so that one atom will undergo the transition to state *B*. The fact that the atom has undergone the transition can be detected in either of two ways. The first is selective field ionization of atoms in state *B* but not those in state *A*, and the second is to detect the optical fluorescence from state *B* but not from state *A*.³⁸ In either case each incoming photon is converted into an ion and an electron or a visible photon, all of which are easily detected. Several variants of this approach have been examined.^{2,39–41}

With optical detection it is unimportant whether state *B* lies above or below state *A*. If *B* lies below *A*, a second approach, triggered superradiance, is possible, and the technique has been carefully examined by Goy *et al.*⁴² To detect the same radiation at ω_{AB} requires an optically thick target of atoms in the upper state *A*, which in a cooled environment would not become superradiant, since spontaneous emission is too slow to initiate the process. However, an incoming photon at frequency ω_{AB} would trigger a superradiant avalanche, converting one incoming photon into $\sim 10^4$ outgoing photons and leaving 10^4 atoms in state *B*, which would be trivial to detect. As shown by the sensitivity given in Table 5.3, this approach is equally as sensitive as the more straightforward approach described above.

While Rydberg atoms have been used to detect infrared radiation with wavelengths as short as $2.34 \mu\text{m}$, they are most effective as detectors of far infrared

Table 5.3. *Far infrared detection sensitivity in Rydberg atom experiments.*

Method	Source	Wavenumber cm^{-1}	NEP $\text{W}/\text{cm}^2\text{Hz}^{1/2}$
Absorption, ^a field ionization	300 K black body	22	10^{-14}
Absorption, ^b field ionization	Far infrared laser	20	5×10^{-15}
Absorption, ^c field ionization	Variable temperature black body	3.3	10^{-17}
Triggered ^d superradiance	Klystron harmonic	3.6	3×10^{-17}

^a(see ref. 2)^b(see ref. 41)^c(see ref. 40)^d(see ref. 42)

radiation, and in Table 5.3 we present measured sensitivities for Rydberg atom detection of far infrared radiation.

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